

12. Weight of first box = 200 kg.

Weight of third box = 125% of 200 kg = 250 kg.

Weight of second box = 120% of 250 kg = 300 kg.

Weight of fourth box = 350 kg.

Let the weight of fifth box be x kg.

Then, 70% of $x = 350$ kg $\Rightarrow x = \left(\frac{350 \times 100}{70}\right) = 500$ kg.

Average weight of four heaviest boxes

$$= \left(\frac{500 + 350 + 300 + 250}{4}\right) \text{ kg} = 350 \text{ kg}.$$

Average weight of four lightest boxes

$$= \left(\frac{200 + 250 + 300 + 350}{4}\right) \text{ kg} = 275 \text{ kg}.$$

$\therefore$ Required difference = $(350 - 275)$ kg = 75 kg.

13. Let Arun's weight be X kg.

According to Arun, $65 < X < 70$.

According to Arun's brother, $60 < X < 70$.

According to Arun's mother, $X \geq 68$ i.e. $X \leq 68$.

The values satisfying all the above conditions are 66, 67 and 68.

$$\therefore \text{Required average} = \left(\frac{66 + 67 + 68}{3}\right) \text{ kg} = \left(\frac{201}{3}\right) \text{ kg} = 67 \text{ kg}.$$

14. Average of 20 numbers = 0.

$\therefore$ Sum of 20 numbers = $(0 \times 20) = 0$.

It is quite possible that 19 of these numbers may be positive and if their sum is a , then 20th number is $(-a)$.

15. Required mean = $\left(\frac{2 + 4 + 6 + \dots + 20}{10}\right) = \frac{2(1 + 2 + \dots + 10)}{10}$
 $= \left(\frac{1}{5} \times \frac{10 \times 11}{2}\right) = 11.$

$$\left[\because 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}\right]$$

16. Required mean = $\left(\frac{1 + 2 + \dots + 11}{11}\right) = \left(\frac{1}{11} \times \frac{11 \times 12}{2}\right) = 6.$

$$\left[\because 1 + 2 + \dots + n = \frac{n(n+1)}{2}\right]$$

17. Average = $\left(\frac{10 + 15 + 20 + 25 + 30}{5}\right) = \frac{100}{5} = 20.$

18. Average = $\frac{3(1 + 2 + 3 + 4 + 5)}{5} = \frac{45}{5} = 9.$

19. Average = $\left(\frac{2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23}{9}\right)$
 $= \frac{100}{9} = 11\frac{1}{9}.$

20. Clearly, we have

$$\left(\frac{3 + 11 + 7 + 9 + 15 + 13 + 8 + 19 + 17 + 21 + 14 + x}{12}\right) = 12$$

or $137 + x = 144$ or $x = 144 - 137 = 7.$

21. We have : $\left(\frac{2 + 7 + 6 + x}{4}\right) = 5$ or $15 + x = 20$ or $x = 5.$

$$\text{Also, } \left(\frac{18 + 1 + 6 + x + y}{5}\right) = 10 \text{ or } 25 + 5 + y = 50 \text{ or } y = 20.$$

22. $P + C + M = C + 120 \Rightarrow P + M = 120.$

$$\therefore \text{Required average} = \frac{P + M}{2} = \frac{120}{2} = 60.$$

23. Required average = $\left(\frac{1 + 2 + 3 + \dots + 100}{100}\right)$
 $= \frac{1}{100} \times \frac{100 \times 101}{2} = 50.5.$

24. Sum of odd numbers upto 100 = $1 + 3 + 5 + \dots + 99$
 $= \frac{50}{2} [2 + (50 - 1) \times 2] = 2500.$

$$\left[\because \text{Sum of } n \text{ terms of an A.P. with first term } a \text{ and common diff. } d = \frac{n}{2} [2a + (n - 1)d]\right]$$

$$\therefore \text{Required average} = \frac{2500}{50} = 50.$$

25. Sum of ages of father and mother = (35×2) years
 $= 70$ years.

Sum of ages of father, mother and son = (27×3) years
 $= 81$ years.

$\therefore$ Son's age = $(81 - 70)$ years = 11 years.

26. $X_1 + X_2 + X_3 = (14 \times 3) = 42.$

$$2(X_2 + X_3) = 30 \Rightarrow X_2 + X_3 = 15.$$

$$\therefore X_1 = (42 - 15) = 27.$$

27. $x_1 + x_2 + x_3 + x_4 = 16 \times 4 = 64.$

$$\frac{1}{2}(x_2 + x_3 + x_4) = 23 \Rightarrow x_2 + x_3 + x_4 = 46.$$

$$\therefore x_1 = 64 - 46 = 18.$$

28. We have : $\left[\frac{x + (x + 2) + (x + 4) + (x + 6) + (x + 8)}{5}\right] = 11$ or

$$5x + 20 = 55 \text{ or } x = 7.$$

So, the numbers are 7, 9, 11, 13, 15.

$$\therefore \text{Required mean} = \left(\frac{11 + 13 + 15}{3}\right) = \frac{39}{3} = 13.$$

29. We have : $\left(\frac{a + b + c}{3}\right) = M$ or $(a + b + c) = 3M.$

$$\text{Now, } (a + b + c)^2 = (3M)^2 = 9M^2$$

$$\Leftrightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 9M^2$$

$$\Leftrightarrow a^2 + b^2 + c^2 = 9M^2. \quad [\because (ab + bc + ca) = 0]$$

$$\therefore \text{Required mean} = \left(\frac{a^2 + b^2 + c^2}{3}\right) = \frac{9M^2}{3} = 3M^2.$$

30. Average = $\left(\frac{11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + 99}{9}\right)$

$$= \left[\frac{(11 + 99) + (22 + 88) + (33 + 77) + (44 + 66) + 55}{9}\right]$$

$$= \left(\frac{4 \times 110 + 55}{9}\right) = \frac{495}{9} = 55.$$